ULTRA-SAFE CENTRAL CAPSULE

Master Document Index & Document Tree

Document Number	01	**Document Type**	Management
Title	Master Document Index & Document Tree	**Revision**	A (Initial Issue)
Status	[PROPOSED] — Concept / Preliminary Maturity	**Date**	2026-09-09
Controlling Baseline	Document 00 Version 3.0 — Comprehensive Engineering Baseline	**Classification**	Internal Engineering

This document establishes the controlled Master Document Index and hierarchical Document Tree for the Ultra-Safe Central Capsule project. It provides the authoritative map of all engineering work products, their parent-child relationships, maturity status, and configuration control status. Document 00 V3.0 remains the single controlling baseline.

1. PURPOSE AND SCOPE

The purpose of this document is to define and maintain the complete set of engineering documents required for the design, analysis, verification and configuration management of the Ultra-Safe Central Capsule. It identifies document numbers, titles, types, maturity levels, parent dependencies and current status.

Scope

This index covers all primary engineering documents from the top-level baseline (Document 00) through system, subsystem, analysis, verification and interface documents. Lower-level drawings, manufacturing instructions and supplier data packages are tracked separately under configuration management and are referenced but not listed in full detail herein.

This document does not authorize design changes. Any proposed deviation from Document 00 V3.0 shall be marked [BASELINE CONFLICT — CHANGE CONTROL REQUIRED] and processed through formal configuration control.

2. APPLICABLE BASELINE / REFERENCES

2.1 Controlling Baseline

Document 00 Version 3.0 — Comprehensive Engineering Baseline is the single controlling authority. All lower-level documents shall comply with or formally change the baseline.

2.2 Locked Baseline Parameters (from Document 00 V3.0)

Parameter	Value	Status
External diameter	5.80 m	**[LOCKED]**
Overall height	3.80 m	**[LOCKED]**
Occupancy	7 persons (pilot + co-pilot + 5)	**[LOCKED]**
Maximum operational mass / hard design ceiling	22,000 kg	**[LOCKED]**
Primary structure	Continuous six-axis CFRP roll-cage with titanium nodes and continuous floor	**[LOCKED]**
Floor hatch	None	**[LOCKED]**
Access	Two side doors + one reinforced roof hatch	**[LOCKED]**
Landing system	8 landing legs with progressive energy absorption	**[LOCKED]**
Recovery concept	8 primary + 8 auxiliary/stabilizer parachutes	**[LOCKED]**
Life-support target	72 h for 7 occupants	**[LOCKED]**
Location / recovery aids	406 MHz COSPAS-SARSAT + GNSS + satellite messaging + visual/acoustic aids	**[LOCKED]**

2.3 Six Functional Survival Layers (Document 00 V3.0)

The safety architecture is defined exclusively by the following six functional survival layers. Technologies may be assigned only when technically justified; independence shall be demonstrated, not assumed from duplication.

1. Primary automatic emergency detection / initiation
2. Independent backup initiation path
3. Independent manual / crew initiation
4. Independent final / last-resort separation path
5. Direct-impact survival
6. Post-event survival and recovery

3. DEFINITIONS AND ABBREVIATIONS

Term	Definition
CFRP	Carbon Fibre Reinforced Polymer
CG	Centre of Gravity
COSPAS-SARSAT	International satellite-based search and rescue system (406 MHz)
Doc 00 V3.0	Controlling Comprehensive Engineering Baseline
FMEA / FTA	Failure Modes and Effects Analysis / Fault Tree Analysis
GNSS	Global Navigation Satellite System
SF	Safety Factor (baseline statement SF ≥ 2.8 is not by itself sufficient)
TBD	To Be Determined — explicit open item requiring closure
VERIFICATION GATE OPEN	Requirement or claim lacks completed verification evidence

4. MASTER DOCUMENT TREE

The document tree is organized by hierarchy. Parent documents control child documents. Maturity levels: CONCEPT → PRELIMINARY → DETAILED → VERIFICATION-READY.

4.1 Top-Level Hierarchy

Doc No.	Title	Type	Maturity	Status	Parent
00	Comprehensive Engineering Baseline	Baseline	Locked	**[LOCKED]**	—
01	Master Document Index & Document Tree (this document)	Management	Preliminary	**[PROPOSED]**	00
02	System Requirements Specification	Requirements	Preliminary	**[TBD]**	00
03	System Architecture & Design Description	Design	Concept	**[TBD]**	00 / 02
04	Mass Properties & CG Control Plan	Design / Analysis	Concept	**[TBD]**	00 / 02
05	Structural Design Description & Load Paths	Design	Concept	**[TBD]**	00 / 03
06	Landing Gear & Energy Absorption System Design	Design	Concept	**[TBD]**	00 / 03
07	Parachute Recovery System Design Description	Design	Concept	**[TBD]**	00 / 03
08	Life Support System Design Description	Design	Concept	**[TBD]**	00 / 03
09	Electrical Power & Avionics Architecture	Design	Concept	**[TBD]**	00 / 03
10	Safety Architecture & Independence Analysis	Safety / Analysis	Concept	**[TBD]**	00
11	FMEA / FTA Master	Safety	Concept	**[TBD]**	10
12	Impact Survival & Energy Management Analysis	Analysis	Concept	**[TBD]**	05 / 06
13	Parachute Deployment & Loads Analysis	Analysis	Concept	**[TBD]**	07
14	Life Support Mass / Energy Balance Analysis	Analysis	Concept	**[TBD]**	08
15	Flotation, Stability & Water Survival Analysis	Analysis	Concept	**[TBD]**	03 / 06
16	Verification & Validation Plan	Verification	Concept	**[TBD]**	02 / 10
17	Interface Control Document (ICD) — Host Platform	Interface	Concept	**[TBD — DEPENDENCY]**	03
18	Configuration Management & Change Control Plan	Management	Preliminary	**[PROPOSED]**	00 / 01

Note: Documents 02–18 are currently placeholders. Content shall be generated only when the corresponding engineering maturity is reached and all parent dependencies are satisfied. No design values beyond Document 00 V3.0 locked parameters are authorized by this index.

5. DOCUMENT TYPE DISCIPLINE

Each document shall contain only content appropriate to its declared type:

- **Requirements** — requirements, rationale, traceability, verification approach. No design solutions.
- **Design** — architecture, design definition, interfaces, calculations, design rationale.
- **Analysis** — assumptions, model, equations, inputs, boundary conditions, results, sensitivity, conclusions.
- **Test** — objectives, article/configuration, instrumentation, procedure, conditions, acceptance criteria, results/status.
- **Verification** — requirement-to-verification mapping and evidence status.
- **Safety (FMEA/FTA)** — failure modes, causes, effects, detection, severity/probability, controls, dependencies.
- **Interface** — physical, structural, electrical, data, thermal, fluid and operational interfaces.
- **Management** — planning, indexing, configuration control, process.

6. TRACEABILITY REQUIREMENTS

Every substantive engineering work product shall maintain bidirectional traceability along the chain:

Source / Need → Requirement → Design Element → Analysis / Implementation → Verification Method → Acceptance Criterion → Evidence → Configuration

If a requirement has no valid source it shall be identified as self-derived with technical rationale. If a design feature has no requirement or engineering justification, the gap shall be identified rather than inventing a requirement.

7. CONFIGURATION AND CHANGE CONTROL

The following items are under strict configuration control and shall not be changed silently:

- Baseline geometry (5.80 m diameter, 3.80 m height)
- Occupancy (7 persons)
- Mass ceiling (22,000 kg)
- Safety architecture (six functional survival layers)
- Structural concept (six-axis CFRP roll-cage with titanium nodes, continuous floor, no floor hatch)
- Access concept (two side doors + reinforced roof hatch)
- Landing concept (8 progressive energy-absorbing legs)
- Recovery concept (8 primary + 8 auxiliary/stabilizer parachutes)
- Life-support duration target (72 h)
- Location/recovery architecture (406 MHz COSPAS-SARSAT + GNSS + satellite messaging + visual/acoustic aids)
- Previously approved requirements

Any proposed change shall be explicitly marked [BASELINE CONFLICT — CHANGE CONTROL REQUIRED] and shall identify affected documents, requirements, interfaces, analyses and verification activities.

8. ASSUMPTIONS / TBDs / OPEN ITEMS

ID	Description	Status	Dependency / Closure Path
OI-01	System Requirements Specification (Doc 02) content	[TBD]	Requires formal derivation from Doc 00 V3.0 needs

OI-02	Host platform class definition and interface requirements	**[TBD — DEPENDENCY]**	Doc 17 ICD; host compatibility must be demonstrated, never assumed
OI-03	Detailed mass budget allocation closing to ≤ 22,000 kg	**[TBD]**	Doc 04 Mass Properties; all budgets must mathematically close
OI-04	Quantitative impact energy allocation among legs, structure, seats	**[TBD]**	Doc 12 Impact Survival Analysis ($E = ½mv^2$ basis)
OI-05	Parachute system sizing (drag area, reefing, opening loads)	**[TBD]**	Doc 13; no final sizing claimed without supporting analysis
OI-06	Life-support mass/energy balances for 72 h / 7 occupants	**[TBD]**	Doc 14; O■, CO■, thermal, power, water, food, leakage
OI-07	Independence demonstration for the six survival layers	**[TBD]**	Doc 10; architectural, physical, electrical, software, common-cause analysis
OI-08	Verification evidence for all locked baseline claims	**[VERIFICATION GATE OPEN]**	Doc 16 V&V; Plan and subsequent evidence packages

9. CONCLUSIONS / STATUS

This Master Document Index & Document Tree (Document 01, Revision A) establishes the controlled framework for all subsequent engineering documentation of the Ultra-Safe Central Capsule. Document 00 Version 3.0 remains the single controlling baseline. All locked parameters and the six functional survival layers are inherited without modification.

Documents 02–18 are identified as required work products. Their content is [TBD] pending satisfaction of parent dependencies and progression through the defined maturity gates. No unverified performance claims, material allowables, test results or certification statements are contained in or authorized by this document.

This document is released at [PROPOSED] status for internal engineering use and configuration management. Formal approval and elevation to [APPROVED] status requires completion of the internal quality gate review against Document 00 V3.0 compliance.

10. DOCUMENT CONTROL RECORD

Rev	Date	Description	Author	Approval
A	2026-09-09	Initial issue — Master Document Index & Document Tree established under Doc 00 V3.0 control	Systems Engineering AI	**[PROPOSED]**

— END OF DOCUMENT 01 —

Ultra-Safe Central Capsule • Controlled Engineering Documentation • Doc 00 V3.0 Baseline